

Does LFP2Vec transfer across labs, and can its confidence be trusted?

An independent reproduction and calibration audit on public data. Keshav Rajput, September 2026. Code, data manifests, exact splits and every number below: <https://github.com/happycoder/lfp2vec-audit>

Question

LFP2Vec (He et al., NeurIPS 2025) fine-tunes an audio model on raw local field potential to predict which brain region an electrode sits in, and reports zero-shot transfer between labs. Its Broader Impact section states that clinical use “should include calibrated uncertainty estimates”. No calibration metric appears in the paper, nor any baseline that uses electrode position rather than signal. This note measures both, on the two public datasets the paper uses, against every control on the same folds.

Setup

Data. Seven IBL insertions from five labs and ten Allen Visual Coding probes from two sessions, chunked exactly as the upstream code does: three seconds, one hundred chunks per channel, five region labels. 132 600 IBL chunks and 50 200 Allen chunks. Two Allen probe files that advertise data and contain only zeros were detected and excluded; IBL contains no CA2 channels at all.

Method. facebook/wav2vec2-base fine-tuned with the paper’s stated hyper-parameters. The self-supervised continuation stage is not run: the upstream code runs it per dataset and never sees two labs, so it cannot be what makes transfer work, and the paper’s own ablation shows it adding 0.000 on IBL. No pretrained weights have been released, so published numbers are read from figures and carry about ± 0.01 .

Evaluation. Leave-one-session-out within lab, all target probes across labs, permutation controls on every fold. Balanced accuracy with chance computed from the classes each test set actually contains, and raw accuracy against the majority-class rate where the paper uses that.

Three results

1. Within lab, the reproduction agrees with the paper, and electrode position beats it. Across three held-out IBL sessions the fine-tune reaches 0.716 balanced accuracy against the paper’s 0.68. Four numbers describing where the contact sits, with the voltage discarded, reach 0.856 on the same sessions. Fine-tuning adds 0.007 over the untrained audio checkpoint. Replacing every input with a surrogate that keeps its power spectrum and destroys its waveform costs the model 0.008: it is a spectral classifier.

2. Across labs, the reproduction collapses, and its confidence does not. Trained on one lab and tested on the other, the model predicts a single class for 95% of chunks at 0.997 confidence, landing below the majority-class rate in both directions. A linear probe recovers which lab a chunk came from at AUC 0.997 from the fine-tuned embeddings, on probes it never saw. Expected calibration error is 0.10 in-lab and 0.56 across labs. A temperature fitted in-lab moves the cross-lab error by 0.05; the target lab would have needed a temperature four and a half times larger, which requires the labels zero-shot transfer claims not to need. No abstention score ranks its own errors. The permuted-label control is four times better calibrated than the trained model: it knows nothing and says so.

3. One preprocessing choice closes the gap, and does not reach position. The two datasets’ spectra agree below 100 Hz and diverge up to 768-fold above 300 Hz, because one pipeline band-passes and the other does not. Low-passing every input at 100 Hz, a corner fixed before any cross-lab number existed, moves the model from below majority to the published margin in both directions, cuts calibration error roughly in half, and drops lab identity to 0.83 and 0.74. With the paper’s own post-processing on top it beats the published margin in one direction and ties it in the other, with no labels from the target lab.

raw accuracy minus majority-class rate	IBL $\rightarrow$ Allen	Allen $\rightarrow$ IBL
paper, Figure 2e (<i>figure</i>)	+0.11	+0.12
reproduction, full band	−0.03	−0.06
harmonised band + paper’s post-processing	+0.15	+0.12
electrode position, no signal	+0.27	+0.21

Two levers were ruled out along the way. The paper’s post-processing, averaging logits over a channel then voting over five neighbours, moves the collapsed model by 0.004. Subtracting each probe’s mean embedding removes the lab signature almost entirely and leaves transfer at chance: the two labs are not the same structure displaced.

What this says

The reduced method does what the paper reports within a lab, does not beat where the electrode is, and transfers across labs only once the labs’ filters are matched. What limits it is not representation but preprocessing the paper

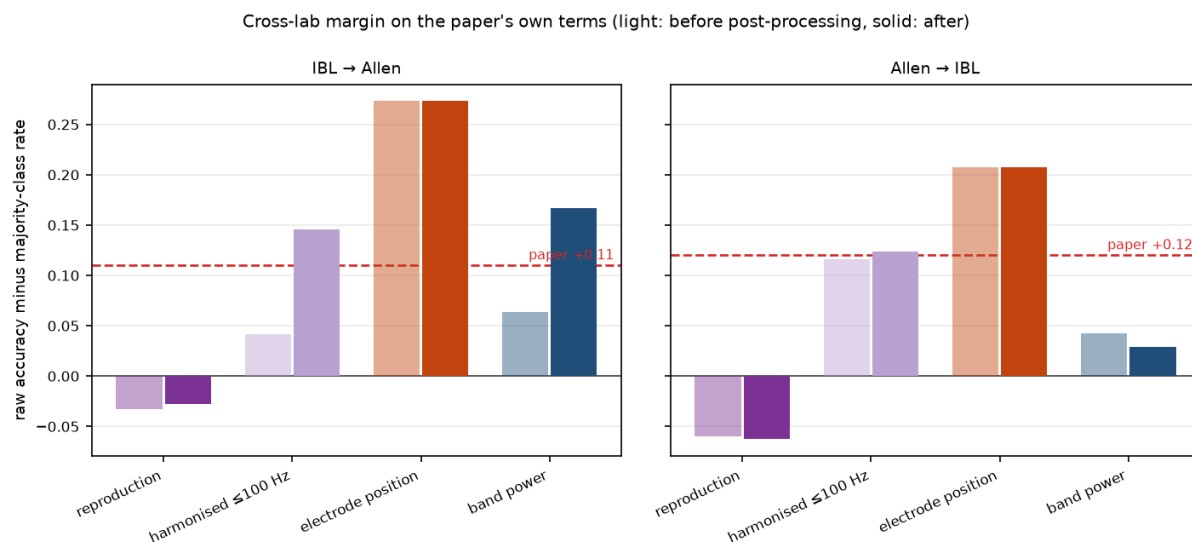


Figure 1: Cross-lab margin on the paper's own terms. Light bars before the paper's post-processing, solid after; dashed line is the published margin.

does not control, and its confidence carries no information about whether it is in a lab it has seen. The measurement the paper calls for and does not make, calibration under shift, is the one that would have revealed this.

One next experiment. Fine-tune with the paper's full pipeline on inputs harmonised *before* self-supervision, and report calibration error beside accuracy. If the lab-identity score stays near 0.8, the residual is physiology; if it falls toward 0.5, the acquisition signature was the whole story.

Limitations

Two of the paper's four datasets are private and untested. Labels are histological estimates taken as given. Three within-lab folds and one seed. Every published number quoted is read from a bitmap. Much of the code was generated with Claude; every experiment, split and number was independently checked, and the repository's development notes say what was.